

Cognitive Modelling: From GOMS to Deep Reinforcement Learning

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ABSTRACT

This course introduces computational cognitive modeling for researchers and practitioners in the field of HCI. Cognitive models use computer programs to model how users perceive, think, and act in human-computer interaction. They offer a powerful approach for understanding interactive tasks and improving user interfaces. This course starts with a review of classic architecture based models such as GOMS and ACT-R. It then rapidly progresses to introducing modern modelling approaches powered by machine learning methods, in particular deep learning, reinforcement learning (RL), and deep RL. The course is built around hands-on Python programming using notebooks.

CCS CONCEPTS

• **Human-centered computing** → *HCI theory, concepts and models.*

KEYWORDS

Cognitive modeling, cognitive architectures, deep learning, reinforcement learning, computational rationality, cooperative intelligence, user interface optimization

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1 INTRODUCTION

Benefits. The course will cover the state-of-the-art in cognitive computational modelling for HCI. The attendees will learn how contemporary modelling paradigms have evolved from the classic approaches used in HCI. They will also get to understand how recent advances in machine learning and cognitive science have made it possible for the state-of-the-art to emerge. After the course, the attendees will be able to approach theoretical and practical HCI problems from the perspective of computational cognitive modelling. They are provided skills and concrete tools with which to apply models in their work, alongside the theoretical basis for how to develop these models further.

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Computational cognitive modelling has emerged – after somewhat of a hiatus – as an important and rapidly advancing paradigm for understanding interaction. Integration of recent advances in machine learning and cognitive science thinking has resulted in models that are able to shed light on such key topics of HCI as multitasking [4], gaze-based selection [2], and touchscreen typing [3]. As these models become more advanced and readily employable in HCI research and design, it is important for everyone – not just the modellers – to understand their theoretical and technical background. The motivation for taking this course is to stay up-to-date with the recent developments and learn to utilise them in one's own work, both theoretical and practical.

Intended audience. The course is primarily targeted at researchers and practitioners in HCI with interest in applications of computational methods and machine learning to the problem of understanding users. Industrial practitioners, interested for example in user modeling, may also find the course valuable.

Prerequisites. While the course will cover advanced topics in machine learning, such as deep learning and reinforcement learning, the audience is not expected to have a machine learning background. The course will provide a practical introduction to these topics, and Python notebooks will be used to interactively learn to use existing models and provide a basis for the attendees to learn to develop new models. The audience is therefore expected to have Python notebooks installed on their computers, and at least an introductory level knowledge of Python and programming in general.

2 CONTENTS

The course will focus on cognitive models driven by modern ML methods and on applying these techniques to concrete HCI problems.

The course consists of 4 x 75 minute sessions:

- Introduction to cognitive modeling
- Deep learning
- Reinforcement learning
- Modeling workflow

Each is expanded in the following subsections:

2.1 Introduction to cognitive modeling

In this session we will critically review the history of cognitive modeling in HCI over the past 50 years. The review will cover modeling approaches that include mathematical models of movement time, procedural models of skill, models of eye movements, computational simulations of human information processing, models inspired by how animals forage for food and models of human learning, rationality and decision making.

The strengths and weaknesses of each approach will be considered with respect to their contribution, both real and perceived, to HCI. The section discusses the uses of models in HCI, such as advancing scientific understanding, evaluating designs, and facilitating cooperative AI. After completing the section the attendees will understand the role of computational cognitive modeling in HCI, and know how the historical approaches have converged into the state-of-the-art of modeling.

2.2 Deep learning models

The session introduces deep learning as an approach to building cognitive models in HCI. Deep learning has been successfully used in modelling various types of interactive behaviour, such as visual search [6] and mid-air interaction in VR [1]. One of the advantages of deep learning is in its ability to discover features and structure in data that are important for explanation and prediction. Deep learning models also excel at processing input data that have a large feature space and contain vast amounts of information. This permits modelers to exploit them, for instance in predicting, from raw pixel data of a UI, user behaviors, without having to extract the visual features of the UI by hand. This section introduces these benefits of deep learning in modeling interaction with recent examples. Python notebooks accompany the lecture of the session, with hands-on practices where the attendees modify existing model code to explore how deep learning models of interaction work. After completing this section, attendees understand the role of deep learning in HCI models, and have experience in applying such models to predict user behavior with realistic UIs.

2.3 Reinforcement and Deep Reinforcement learning

The session provides an introduction to creating RL based simulation models of interactive tasks. The module covers basic tabular methods (temporal difference RL) and advanced, deep RL approaches, with the focus on how these techniques have been used in recent breakthrough models in HCI and machine learning. We discuss the theoretical underpinnings of RL as the basis of simulating human adaptation in interactive tasks. The session covers modeling of such phenomena as multitasking, typing, and visual search. After completing this section, attendees will be able to read and understand research papers in HCI that make use of RL based computational cognitive models, make use of such models in their work, and participate in designing them further.

2.4 How to build a model: Modeling workflow

How to construct a model for a practical purpose? In this unit, we adapt the modern modeling workflow [5] to the case of RL-based cognitive modeling in HCI. We focus on five topics: (1) modeling human cognitive capabilities using RL, (2) model testing (e.g., parameter recovery checking), (3) creation of testing and training environments for RL agents, (4) parameter inference, and (5) validation against benchmark datasets in HCI. After completing this section, attendees will have necessary tools for model construction and validation.

2.5 Practical work

The course makes heavy use of Python notebooks. Each session includes a notebook that has been prepared for that session, with activities throughout the session carried out on the notebooks. The notebook platform provides an easy way for the attendees to follow the session and explore and experiment on relevant model code. Each session includes short assignments that have the attendees carry out tasks using the notebooks. The goal of this interactivity is that the attendees get hands-on experience with computational cognitive models, not just theoretical overview and examples.

3 INSTRUCTOR BACKGROUND

Jussi P.P. Jokinen is an Assistant Professor of cognitive science at University of Jyväskylä. He was previously a Research Fellow at Finnish Center of AI FCAI. His doctorate is from cognitive science in University of Jyväskylä in 2015. His award-winning work on computational cognitive modelling has been published in multiple HCI and cognitive science venues, such as ACM CHI, IJHCS, Human Factors, Cognitive Science.

Antti Oulasvirta is Professor of user interfaces at Aalto University where he leads the Interacti AI research programme at the Finnish Center for AI. He was previously a Senior Researcher at the Max Planck Institute for Informatics. He received his doctorate in Cognitive Science from the University of Helsinki in 2006, after which he was a Fulbright Scholar at the School of Information in University of California-Berkeley in and a Senior Researcher at Helsinki Institute for Information Technology HIIT. He was awarded an ERC Starting Grant (2015-2020) for research on computational design of user interfaces.

Andrew Howes is Professor of computer science at University of Birmingham. He was previously FCAI Visiting Professor at Aalto University, Marshall M. Weinberg, Visiting Professor, Department of Psychology, University of Michigan in 2016 and visiting professor at NASA Ames Research Center in 2002 and 2015. Howes is a leader in computational rationality. Howes's interdisciplinary work has been published in the most prestigious journals in Cognitive Science and in HCI and 5 of his papers have won honorable mentions at ACM CHI. He has been associate editor at Cognitive Science journal and International Journal of Human-Computer Studies. He was papers chair of the 2017 33rd International Meeting of the Cognitive Science Society and is a former Head of Computer Science at the University of Birmingham.

4 RESOURCES

Projects and other material by the instructors that take use of computational cognitive modelling. Some of these models will also be employed in the course.

- <https://userinterfaces.aalto.fi/touchscreen-typing/>
- <https://users.aalto.fi/~jokinej10/visual-search/>
- <https://github.com/aditya02acharya/TypingAgent>
- <https://gitlab.com/jokinenj/multitasking-driving>

5 ACCESSIBILITY

The course makes heavy use of Python notebooks, which fortunately offer a large range of accessibility tools. For instance, the

Accessibility Toolbar (AT4N, <https://pypi.org/project/accessibility-toolbar/>) supports dyslexic and visually impaired users.

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